MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **WEEKLY CHAPTER TEST – 03 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. If one zero of the polynomial $f(x) = (k^2+4)x^2 + 13x + 4k$ is reciprocal of the other, then $k =$: [1]

(A) 2 (B) -2 (C) 1 (D) -1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Let the zeroes of $f(x)$ be α and $\frac{1}{\alpha}$ $(\alpha \neq 0)$. We know the product of zeroes relation:

Formula Used: Product of zeroes $= \alpha\beta = \frac{c}{a}$

Here $a = k^2+4$ and $c = 4k$, so:

$$\alpha \cdot \frac{1}{\alpha} = \frac{4k}{k^2+4} \implies 1 = \frac{4k}{k^2+4} \implies k^2 - 4k + 4 = 0$$
$$\implies (k-2)^2 = 0 \implies \boxed{k = 2}. \quad [1M]$$

Q2. If α and β are the zeroes of the polynomial $f(x) = x^2 + x - 1$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is equal to: [1]

(A) 1 (B) -1 (C) 0 (D) 2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

From $f(x) = x^2 + x - 1$, we have $a = 1$, $b = 1$, $c = -1$. We know Vieta's formulas:

Formula Used: $\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$

Thus $\alpha + \beta = -1$ and $\alpha\beta = -1$. Therefore:

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-1}{-1} = \boxed{1}. \quad [1M]$$

Q3. If one zero of the polynomial $f(x) = 5x^2 + 13x + k$ is reciprocal of the other, then the value of k is: [1]

(A) 0 (B) 5 (C) $\frac{1}{6}$ (D) 6

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B)

Let the zeroes be α and $\frac{1}{\alpha}$. We know that for $ax^2 + bx + c$:

Formula Used: $\alpha\beta = \frac{c}{a}$

Comparing $5x^2 + 13x + k$, we have $a = 5$ and $c = k$. Hence:

$$\alpha \cdot \frac{1}{\alpha} = \frac{k}{5} \implies 1 = \frac{k}{5} \implies \boxed{k = 5}. \quad [1M]$$

Q4. The graph of $y = p(x)$ intersects the x-axis at $(-3, 0)$ and $(5, 0)$. The number of zeroes of $p(x)$ is: [1]

(A) 0 (B) 1 (C) 2 (D) 3

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (C)

We know the geometric meaning of the zeroes of a polynomial $p(x)$:

Formula Used: Number of real zeroes of $p(x)$ = Number of points where $y = p(x)$ cuts/touches the x-axis

Since $y = p(x)$ intersects the x-axis at exactly 2 distinct points, namely $(-3, 0)$ and $(5, 0)$, the number of zeroes of $p(x)$ is $\boxed{2}$ (which are $x = -3$ and $x = 5$). [1M]

Q5. Assertion-Reason: [1]

Assertion (A): If the zeroes of $ax^2 + bx + c$ are equal in magnitude but opposite in sign, then $b = 0$.

Reason (R): Sum of zeroes is $-\frac{b}{a}$, and $\alpha + (-\alpha) = 0 \implies -\frac{b}{a} = 0 \implies b = 0$.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) Both A and R are true and R is the correct explanation of A.

We know that for any quadratic polynomial $p(x) = ax^2 + bx + c$ $(a \neq 0)$:

Formula Used: $\alpha + \beta = -\frac{b}{a}$

If $\beta = -\alpha$, then $\alpha + (-\alpha) = 0 \implies -\frac{b}{a} = 0 \implies b = 0$ (since $a \neq 0$). Reason **(R)** accurately states the exact algebraic derivation of Assertion **(A)**. Thus both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**. [1M]

Q6. If α, β are the zeroes of the polynomial $2y^2 + 7y + 5$, write the value of $\alpha + \beta + \alpha\beta$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $2y^2 + 7y + 5$ with $ay^2 + by + c$, we get $a = 2$, $b = 7$, and $c = 5$. We know Vieta's relations:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \qquad \alpha\beta = \dfrac{c}{a}$

Calculating the sum and product of zeroes:

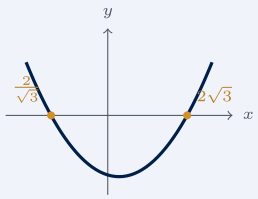
$$\alpha + \beta = -\frac{7}{2}, \qquad \alpha\beta = \frac{5}{2}. \qquad \text{[1M]}$$

Adding the two results:

$$\alpha + \beta + \alpha\beta = -\frac{7}{2} + \frac{5}{2} = \frac{-7+5}{2} = \frac{-2}{2} = \boxed{-1}. \qquad \text{[1M]}$$

Q7. Find the zeroes of the quadratic polynomial $f(x) = \sqrt{3}\,x^2 - 8x + 4\sqrt{3}$ and verify the relationship between the zeroes and the coefficients. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Graphical Verification: Since leading coefficient $a = \sqrt{3} > 0$, the parabola opens upward and cuts the x-axis at $x = \frac{2}{\sqrt{3}}$ and $x = 2\sqrt{3}$.

Comparing $f(x) = \sqrt{3}\,x^2 - 8x + 4\sqrt{3}$ with $ax^2 + bx + c$, we have $a = \sqrt{3}$, $b = -8$, $c = 4\sqrt{3}$. We know:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \qquad \alpha\beta = \dfrac{c}{a}$

Step 1 – Factorisation: Product $a \cdot c = \sqrt{3}(4\sqrt{3}) = 12$. Splitting $-8 = -6 - 2$ (as $(-6)(-2) = 12$):

$$\begin{aligned} f(x) &= \sqrt{3}\,x^2 - 6x - 2x + 4\sqrt{3} \\ &= \sqrt{3}\,x(x - 2\sqrt{3}) - 2(x - 2\sqrt{3}) = (\sqrt{3}\,x - 2)(x - 2\sqrt{3}). \qquad \text{[1M]} \end{aligned}$$

Setting $f(x) = 0$ gives the zeroes $\alpha = \boxed{\dfrac{2}{\sqrt{3}} \text{ or } \dfrac{2\sqrt{3}}{3}}$ and $\beta = \boxed{2\sqrt{3}}$. [1M]

Step 2 – Verification:

$$\alpha + \beta = \frac{2}{\sqrt{3}} + 2\sqrt{3} = \frac{2+6}{\sqrt{3}} = \frac{8}{\sqrt{3}}, \qquad -\frac{b}{a} = -\frac{(-8)}{\sqrt{3}} = \frac{8}{\sqrt{3}}, \qquad \text{[0.5M]}$$

$$\alpha\beta = \left(\frac{2}{\sqrt{3}}\right)(2\sqrt{3}) = 4, \qquad \frac{c}{a} = \frac{4\sqrt{3}}{\sqrt{3}} = 4. \qquad \text{[0.5M]}$$

Q8. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - px + q$, prove that: $\dfrac{\alpha^2}{\beta^2} + \dfrac{\beta^2}{\alpha^2} = \dfrac{p^4}{q^2} - \dfrac{4p^2}{q} + 2$. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

For $f(x) = x^2 - px + q$, we have $a = 1$, $b = -p$, $c = q$ (with $q \neq 0$ for $\alpha, \beta \neq 0$). We know Vieta's formulas and the square identity:

Formula Used: $\alpha + \beta = p, \quad \alpha\beta = q, \qquad x^2 + y^2 = (x + y)^2 - 2xy$

First compute $\alpha^2 + \beta^2$:

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = p^2 - 2q. \qquad \textbf{[1M]}$$

Now simplify the LHS using common denominator $\alpha^2\beta^2$:

$$\begin{aligned} \text{LHS} = \frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2} &= \frac{\alpha^4 + \beta^4}{\alpha^2\beta^2} = \frac{(\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2}{(\alpha\beta)^2} \qquad \textbf{[1M]} \\ &= \frac{(p^2 - 2q)^2 - 2q^2}{q^2} = \frac{p^4 - 4p^2q + 4q^2 - 2q^2}{q^2} \\ &= \frac{p^4 - 4p^2q + 2q^2}{q^2} = \boxed{\frac{p^4}{q^2} - \frac{4p^2}{q} + 2 = \text{RHS}.} \qquad \textbf{[1M]} \end{aligned}$$

Q9. If α and β are the zeroes of the polynomial $x^2 - 6x + k$ such that $3\alpha + 2\beta = 20$, then find the value of k. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Comparing $x^2 - 6x + k$ with $ax^2 + bx + c$, we have $a = 1$, $b = -6$, $c = k$. We know Vieta's relations:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \qquad \alpha\beta = \dfrac{c}{a}$

Thus:

$$\alpha + \beta = 6 \quad \text{— (1)}, \qquad \text{and given} \quad 3\alpha + 2\beta = 20 \quad \text{— (2)}. \qquad \textbf{[1M]}$$

Multiplying equation (1) by 2 gives $2\alpha + 2\beta = 12$. Subtracting this from (2):

$$\begin{aligned} (3\alpha + 2\beta) - (2\alpha + 2\beta) = 20 - 12 &\implies \alpha = 8. \\ \text{From (1):}\ 8 + \beta = 6 &\implies \beta = -2. \qquad \textbf{[1M]} \end{aligned}$$

Using the product of zeroes relation $\alpha\beta = k$:

$$k = \alpha\beta = (8)(-2) = \boxed{-16}. \qquad \textbf{[1M]}$$

Q10. Case Study: High-Diving Athlete Annie

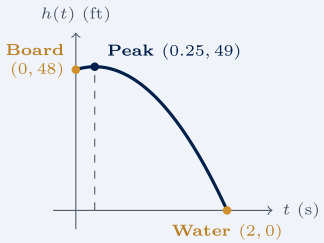
Annie dives from a diving platform. Her height h (in ft) above water t seconds after leaving the board is modelled by $h(t) = -16t^2 + 8t + 48$. [4]

(i) What was Annie's initial height above the water level at the instant she jumped ($t = 0$)? **[1]**

(ii) At what time t will Annie hit the surface of the water in the pool? **[1]**

(iii) **(a)** A second diver, Rita, dives with height polynomial $p(t)$ having zeroes -1 and 2, and initial height 32 ft. Find $p(t)$. **OR (b)** Determine the maximum clearance height achieved by Annie during her dive. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Projectile Trajectory: Annie leaves the diving board at $t = 0$ from height 48 ft, reaches her peak of 49 ft at $t = 0.25$ s, and enters the pool ($h = 0$) at $t = 2$ s.

Formula Used: $h(t) = -16t^2 + 8t + 48, \qquad p(t) = k(t - \alpha)(t - \beta)$

(i) Initial height: Substituting $t = 0$ into $h(t)$:

$$h(0) = -16(0)^2 + 8(0) + 48 = \boxed{48 \text{ ft}}. \qquad [1M]$$

(ii) Time to hit water ($h = 0$): Solving $-16t^2 + 8t + 48 = 0$, divide through by -8:

$$2t^2 - t - 6 = 0 \implies 2t^2 - 4t + 3t - 6 = 0 \implies (2t + 3)(t - 2) = 0 \implies t = -\frac{3}{2} \text{ or } t = 2.$$

Rejecting negative time $t = -\frac{3}{2} < 0$, Annie hits the water at $t = \boxed{2 \text{ s}}$. [1M]

(iii)(a) Rita's dive polynomial $p(t)$: With zeroes $\alpha = -1$, $\beta = 2$:

$$p(t) = k(t + 1)(t - 2) = k(t^2 - t - 2). \qquad [1M]$$

Using initial height $p(0) = 32$: $k(0 + 1)(0 - 2) = 32 \implies -2k = 32 \implies k = -16$. Thus:

$$p(t) = -16(t^2 - t - 2) = \boxed{-16t^2 + 16t + 32}. \qquad [1M]$$

OR (Alternative Part (b) – Annie's maximum height): Completing the square in $h(t)$:

$$\begin{aligned} h(t) &= -16\left(t^2 - \frac{1}{2}t\right) + 48 \\ &= -16\left[\left(t - \frac{1}{4}\right)^2 - \frac{1}{16}\right] + 48 \qquad [1M] \\ &= -16\left(t - \frac{1}{4}\right)^2 + 1 + 48 = -16\left(t - \frac{1}{4}\right)^2 + 49. \end{aligned}$$

Since $-16\left(t - \frac{1}{4}\right)^2 \le 0$, the maximum height attained is $\boxed{49 \text{ ft}}$ (at $t = 0.25$ s). [1M]

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